Chem 150 Answer Key Problem Set 2

1. Complete the following phrases:

Ionization energy _**decreases**__ from the top to the bottom in a group.

Electron affinity __**increases**___from the left to the right within a period.

Atomic radius ____**increases** __ from the top to the bottom in a group.

Electronegativity _ **increases** __ from the left to the right within a period.

The metallic character of an element _ **decreases** _ from the left to the right within a period.

Cations are _____**smaller**____ than the atoms from which they form.

In isoelectronic anions the radius _ **increases** __ as the charge increases (becomes more negative).

2. What is the difference between Electron affinity and electronegativity of an element?

Electronegativity is the ability of an atom *in a compound* to attract shared electrons to itself. Electron affinity is an atomic property (measures the energy change when an *isolated gaseous atom* gains an electron).

3. Write an equation for the third ionization energy of tin.

Sn^{2+} + → Sn^{3+} + e^-

4. What is the effective nuclear charge in a nitrogen atom?

The effective nuclear charge is the portion of the nuclear charge that the valence electrons effectively experience due to the screening of the inner electrons (the core electrons).

Zeff = atomic number - # of inner electrons

5. How many Valence electrons do the following elements possess: He, Si, Br, K

He (2), Si (4), Br (7), K (1)

6. Arrange the following ions in order of increasing size: F^-, O^{2-}, Al^{3+} and Na^+

$Al^{3+} < Na^+ < F^- < O^{2-}$ (all ions have 8 VE, so the greater the number of protons in the nucleus, the greater the attraction of the electrons, the smaller the radius)

7. Using the periodic table arrange the following in the expected order of increasing first ionization energy: B, Li, N, K

K < Li < B < N

8. Arrange the following atoms in order of increasing atomic radii: Mg, Si, Sr, S

S < Si < Mg < Sr

9. Using the periodic table alone, predict the formulas for:

a) the chloride of tin $SnCl_4$

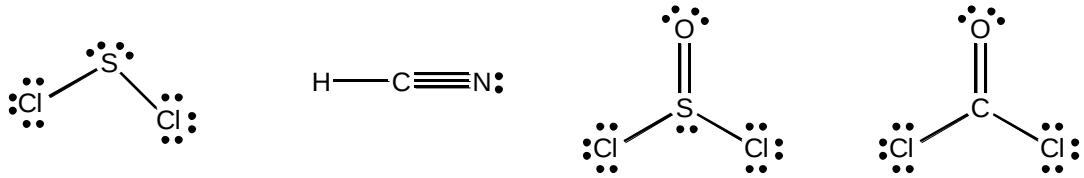
b) the oxide of indium In_2O_3

c) the sulfate of barium $BaSO_4$

d) the hydride of phosphorus PH_3

10. Draw the Lewis symbols (Lewis structures) of the following species: Mg, C, F, S^{2-},O_2

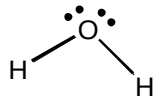
11. Draw Lewis structures of the following molecules: SCl_2, HCN, $SOCl_2$ (sulfur is the central atom) $COCl_2$ (carbon is the central atom)



12. Draw Lewis structures for the following species. Show lone pairs as pairs of dots and show any non-zero formal charges:

_____

H_2O



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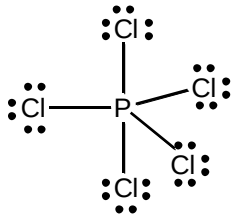
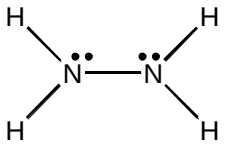
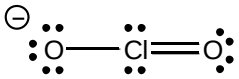


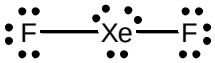
PCl_5



N_2H_4 (The molecule has a N-N single bond)



$[ClO_2]^-$



XeF_2

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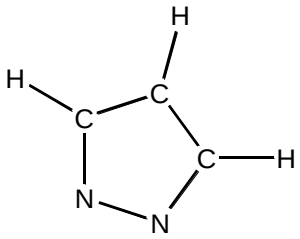
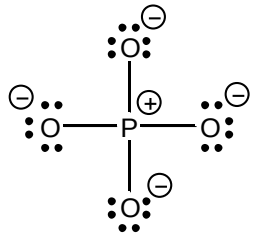
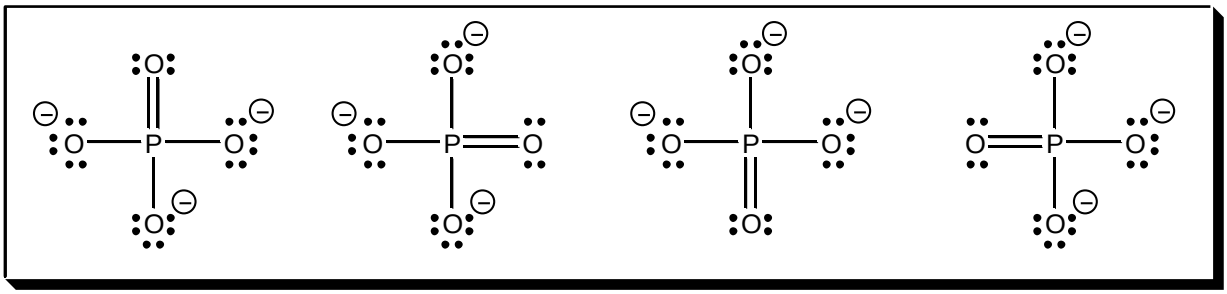
13. Draw all possible Lewis structures for the phosphate ion and the pyrazolate ion **$[C_3H_3N_2]^-$** (see skeletal structure below). Indicate in each case which Lewis structures are equivalent and which are the best Lewis structures.

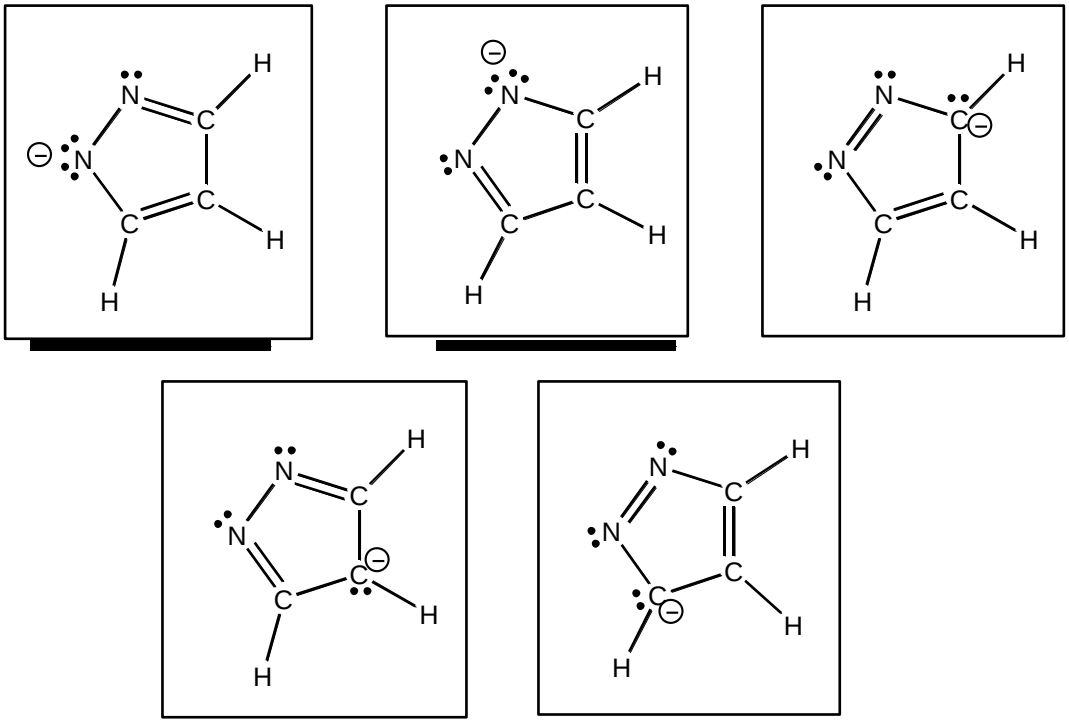


Skeletal structure of the pyrazolate





The best Lewis structures are the framed four structures (also equivalent). The Lewis structure without any double bonds obeys the octet for all atoms but has to many formal charges.



The two underlined Lewis structures are the best Lewis structures since the negative formal charge is on the most electronegative atom (nitrogen). The two best Lewis structures are also equivalent Lewis structures, so are the two structures on the right (top right and bottom right)

14. Hydrogen peroxide (atom connectivity H-O-O-H) is a reactive substance that usually does not exceed 30 % vol. H_2O_2 in aqueous solution commercially available. Pure hydrogen peroxide has rocket fuel properties. Using the bond energy values from the table provided at the end of the chapter 2 handout evaluate which of the two reaction pathways given below is responsible for the instability of this molecule. Estimate the energy change that occurs with each decomposition reaction.

$$\mathbf{H_2O_2 \rightarrow H_2 + O_2}$$

$$\mathbf{2H_2O_2 \rightarrow 2H_2O + O_2}$$

Let's rewrite the equations a bit so we can see the bonds that need to be broken and the bonds that form more easily starting with the top equation:

H-O-O-H → H-H + O=O

We need to break two O-H bonds and one O-O bond which will cost 2 x 464 kJ/mol + 142 kJ/mol = 1070 kJ/mol to produce two hydrogen atoms and two oxygen atoms. We get energy back by forming one H-H and one O=O bond as summarized in the table bellow.

Energy required for bond breaking:	2 mol H-O bonds we need	2 x 464 kJ
	1 mol O-O bonds we need	142 kJ
Total bond breaking energy required to atomize one mol of H_2O_2:		**1070 kJ**
Energy released in bond forming:	1 mol H-H bonds we get	-436 kJ
	1 mol of O=O bonds we get	-498 kJ
Total energy released in forming 1 mol H_2 and 1 mol O_2 from isolated atoms:		**-934 kJ**
Energy change that occurs in the reaction **$H_2O_2 \rightarrow H_2 + O_2$**		**+136 kJ**

This means 135 kJ are required to decompose one mol of hydrogen peroxide in to the elements. This reaction is endothermic.

Let's now consider the decomposition in to water and oxygen:

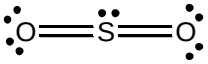
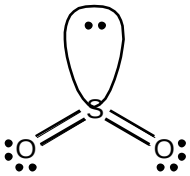
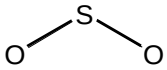
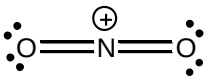
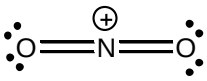
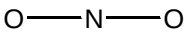
2H-O-O-H →2H-O-H + O=O

Energy required for bond breaking:	4 mol H-O bonds we need	4 x 464 kJ
	2 mol O-O bonds we need	2 x 142 kJ
Total bond breaking energy required to atomize two mol of H_2O_2:		**2140 kJ**

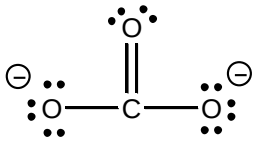
Energy released in bond forming:	4 mol HO bonds we get	4 x-464 kJ
	1 mol of O=O bonds we get	-498 kJ
Total energy released in forming 2 mol H_2O and 1 mol O_2 from isolated atoms:		-2354 kJ
Energy change that occurs in the reaction **$2H_2O_2 \rightarrow 2H_2O + O_2$**		**-214 kJ**

This means 214 kJ are released if two mol of hydrogen peroxide decompose in to water and oxygen. This reaction is exothermic and we would suggest that this is the pathway by which hydrogen peroxide decomposes. This is indeed the observed reaction. Hydrogen peroxide solutions decompose into oxygen and water and are generally stored in the fridge to slow down the decomposition.

15.

Lewis Structure	EGG	Parent shape	molecular shape	polar (Yes or No)
	**3**	 trigonal planar	 bent	Yes
SO_2				
$[NO_2]^+$				
	**2**	 linear	 linear	No

$[CO_3]^{2-}$



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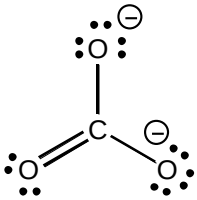
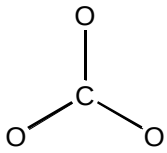


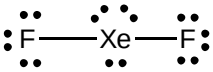
trigonal planar



trigonal planar

No

XeF_2



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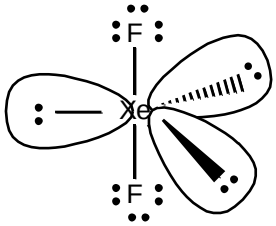
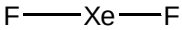


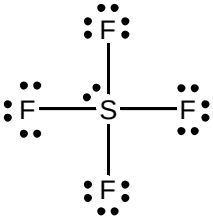
trigonal bipyramidal



linear

No

SF_4



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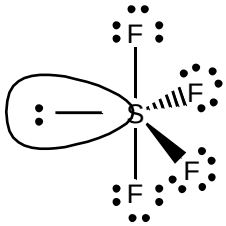
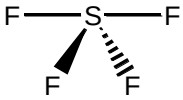


trigonal bipyramidal



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Yes